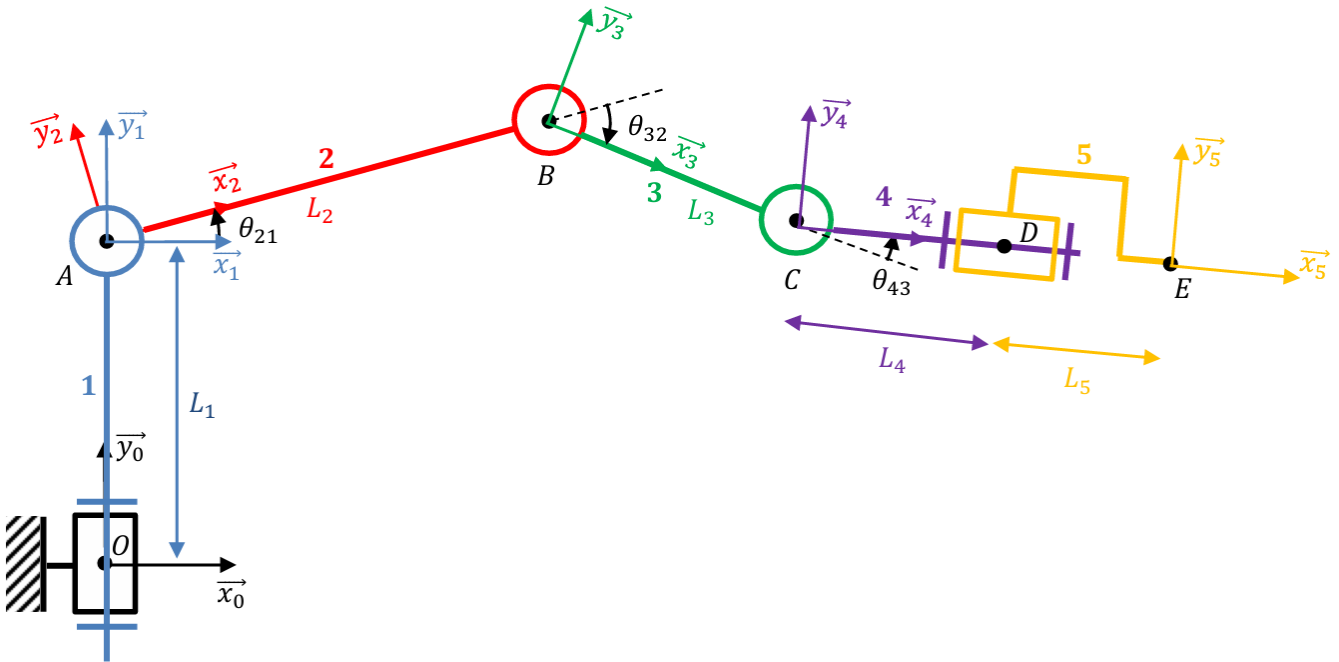


EX 3 : Correction

Bras manipulateur ERICC 3



Cas général

Question 1: Exprimer le vecteur position du point E par rapport au bâti.

$$\begin{aligned}\overrightarrow{OE} &= \overrightarrow{OA} + \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DE} \\ \overrightarrow{OE} &= \lambda_{10}\overrightarrow{y_0} + L_2\overrightarrow{x_2} + L_3\overrightarrow{x_3} + (L_4 + L_5)\overrightarrow{x_4}\end{aligned}$$

Question 2: Exprimer les différents vecteurs rotation du système.

$\overrightarrow{\Omega_{10}}$	$\overrightarrow{\Omega_{21}}$	$\overrightarrow{\Omega_{32}}$	$\overrightarrow{\Omega_{43}}$	$\overrightarrow{\Omega_{54}}$
$\dot{\theta}_{10}\overrightarrow{y_1}$	$\dot{\theta}_{21}\overrightarrow{z_1}$	$\dot{\theta}_{32}\overrightarrow{z_1}$	$\dot{\theta}_{43}\overrightarrow{z_1}$	$\dot{\theta}_{54}\overrightarrow{x_5}$

Question 3: Déterminer la vitesse $\vec{V}(E, 5/0)$ par la définition

$$\vec{V}(E, 2/0) = \frac{d\overrightarrow{OE}}{dt}\bigg|_0$$

$$\vec{V}(E, 2/0) = L_2\overrightarrow{\Omega_{20}} \wedge \overrightarrow{x_2} + L_3\overrightarrow{\Omega_{30}} \wedge \overrightarrow{x_3} + \dot{\lambda}_{54}\overrightarrow{x_4} + (L_4 + L_5)\overrightarrow{\Omega_{50}} \wedge \overrightarrow{x_5}$$

$$\begin{aligned}\vec{V}(E, 2/0) &= L_2(\dot{\theta}_{21}\overrightarrow{z_1} + \dot{\theta}_{10}\overrightarrow{y_1}) \wedge \overrightarrow{x_2} + L_3((\dot{\theta}_{32} + \dot{\theta}_{21})\overrightarrow{z_1} + \dot{\theta}_{10}\overrightarrow{y_1}) \wedge \overrightarrow{x_3} \\ &\quad + (L_4 + L_5)((\dot{\theta}_{43} + \dot{\theta}_{32} + \dot{\theta}_{21})\overrightarrow{z_1} + \dot{\theta}_{10}\overrightarrow{y_1}) \wedge \overrightarrow{x_5}\end{aligned}$$

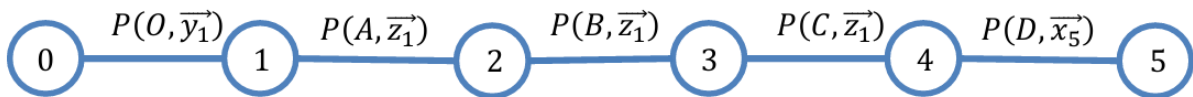
$$\vec{V}(E, 2/0) = (L_2 \dot{\theta}_{21} \vec{z}_1 \wedge \vec{x}_2 + L_2 \dot{\theta}_{10} \vec{y}_1 \wedge \vec{x}_2) + (L_3 (\dot{\theta}_{32} + \dot{\theta}_{21}) \vec{z}_1 \wedge \vec{x}_3 + L_3 \dot{\theta}_{10} \vec{y}_1 \wedge \vec{x}_3) \\ + ((L_4 + L_5) (\dot{\theta}_{43} + \dot{\theta}_{32} + \dot{\theta}_{21}) \vec{z}_1 \wedge \vec{x}_5 + (L_4 + L_5) \dot{\theta}_{10} \vec{y}_1 \wedge \vec{x}_5)$$

$$\begin{aligned} \vec{y}_1 \wedge \vec{x}_2 &= -\cos \theta_{21} \vec{z}_1 \\ \vec{y}_1 \wedge \vec{x}_3 &= -\cos(\theta_{32} + \theta_{21}) \vec{z}_1 \\ \vec{y}_1 \wedge \vec{x}_5 &= -\cos(\theta_{43} + \theta_{32} + \theta_{21}) \vec{z}_1 \\ \vec{z}_1 \wedge \vec{x}_2 &= \vec{z}_2 \wedge \vec{x}_2 = \vec{y}_2 \\ \vec{z}_1 \wedge \vec{x}_3 &= \vec{z}_3 \wedge \vec{x}_3 = \vec{y}_3 \\ \vec{z}_1 \wedge \vec{x}_5 &= \vec{z}_4 \wedge \vec{x}_4 = \vec{y}_4 \quad ; \quad \text{Attention } \vec{z}_4 \neq \vec{z}_5 \end{aligned}$$

$$\vec{V}(E, 2/0) = (L_2 \dot{\theta}_{21} \vec{y}_2 - L_2 \dot{\theta}_{10} \cos \theta_{21} \vec{z}_1) + (L_3 (\dot{\theta}_{32} + \dot{\theta}_{21}) \vec{y}_3 - L_3 \dot{\theta}_{10} \cos(\theta_{32} + \theta_{21}) \vec{z}_1) \\ + ((L_4 + L_5) (\dot{\theta}_{43} + \dot{\theta}_{32} + \dot{\theta}_{21}) \vec{y}_4 - (L_4 + L_5) \dot{\theta}_{10} \cos(\theta_{43} + \theta_{32} + \theta_{21}) \vec{z}_1)$$

$$\vec{V}(E, 2/0) = \mathbf{L_2 \dot{\theta}_{21} \vec{y}_2} + \mathbf{L_3 (\dot{\theta}_{32} + \dot{\theta}_{21}) \vec{y}_3} + \mathbf{(L_4 + L_5) (\dot{\theta}_{43} + \dot{\theta}_{32} + \dot{\theta}_{21}) \vec{y}_4} \\ - \mathbf{\dot{\theta}_{10} (L_2 \cos \theta_{21} + (L_4 + L_5) \cos(\theta_{43} + \theta_{32} + \theta_{21}) + L_3 \cos(\theta_{32} + \theta_{21})) \vec{z}_1}$$

Question 4: Etablir le graphe des liaisons du système.



Question 5: Déterminer la vitesse $\vec{V}(E, 5/0)$ par composition du mouvement

$$\vec{V}(E, 5/0) = \vec{V}(E, 5/4) + \vec{V}(E, 4/3) + \vec{V}(E, 3/2) + \vec{V}(E, 2/1) + \vec{V}(E, 1/0)$$

$$\vec{V}(E, 5/4) = \vec{V}(D, 5/4) + \vec{ED} \wedge \vec{\Omega}_{54} = -L_5 \vec{x}_5 \wedge \dot{\theta}_{54} \vec{x}_5 = \vec{0}$$

$$\begin{aligned} \vec{V}(E, 4/3) &= \vec{V}(C, 4/3) + \vec{EC} \wedge \vec{\Omega}_{43} \\ &= -(L_4 + L_5) \vec{x}_5 \wedge \dot{\theta}_{43} \vec{z}_4 \\ &= \dot{\theta}_{43} (L_4 + L_5) \vec{y}_4 \\ \text{Attention : } \vec{z}_4 &\neq \vec{z}_5 \end{aligned}$$

$$\begin{aligned} \vec{V}(E, 3/2) &= \vec{V}(B, 3/2) + \vec{EB} \wedge \vec{\Omega}_{32} \\ &= -[(L_4 + L_5) \vec{x}_5 + L_3 \vec{x}_3] \wedge \dot{\theta}_{32} \vec{z}_3 \\ &= \dot{\theta}_{32} (L_4 + L_5) \vec{y}_4 + \dot{\theta}_{32} L_3 \vec{y}_3 \end{aligned}$$

$$\begin{aligned} \vec{V}(E, 2/1) &= \vec{V}(B, 2/1) + \vec{EB} \wedge \vec{\Omega}_{21} \\ &= -[(L_4 + L_5) \vec{x}_5 + L_3 \vec{x}_3 + L_2 \vec{x}_2] \wedge \dot{\theta}_{21} \vec{z}_2 \\ &= \dot{\theta}_{21} (L_4 + L_5) \vec{y}_4 + \dot{\theta}_{21} L_3 \vec{y}_3 + \dot{\theta}_{21} L_2 \vec{y}_2 \end{aligned}$$

$$\begin{aligned} \vec{V}(E, 1/0) &= \vec{V}(A, 1/0) + \vec{EA} \wedge \vec{\Omega}_{10} \\ &= -[(L_4 + L_5) \vec{x}_5 + L_3 \vec{x}_3 + L_2 \vec{x}_2 + \lambda_{10} \vec{y}_0] \wedge \dot{\theta}_{10} \vec{y}_1 \end{aligned}$$

$$= -(\dot{\theta}_{10}(L_4 + L_5) \cos(\theta_{43} + \theta_{32} + \theta_{21}) + \dot{\theta}_{10}L_3 \cos(\theta_{32} + \theta_{21}) + \dot{\theta}_{10}L_2 \cos(\theta_{21}))\vec{z}_1$$

$$\begin{aligned}\vec{V}(E, 5/0) &= \dot{\theta}_{43}(L_4 + L_5)\vec{y}_4 + \dot{\theta}_{32}(L_4 + L_5)\vec{y}_4 + \dot{\theta}_{32}L_3\vec{y}_3 + \dot{\theta}_{21}(L_4 + L_5)\vec{y}_4 + \dot{\theta}_{21}L_3\vec{y}_3 + \dot{\theta}_{21}L_2\vec{y}_2 \\ &\quad - (\dot{\theta}_{10}(L_4 + L_5) \cos(\theta_{43} + \theta_{32} + \theta_{21}) + \dot{\theta}_{10}L_3 \cos(\theta_{32} + \theta_{21}) \\ &\quad + \dot{\theta}_{10}L_2 \cos(\theta_{21}))\vec{z}_1\end{aligned}$$

$$\begin{aligned}\vec{V}(E, 5/0) &= (\dot{\theta}_{43} + \dot{\theta}_{32} + \dot{\theta}_{21})(L_4 + L_5)\vec{y}_4 + (\dot{\theta}_{32} + \dot{\theta}_{21})L_3\vec{y}_3 + \dot{\theta}_{21}L_2\vec{y}_2 \\ &\quad - \dot{\theta}_{10}((L_4 + L_5) \cos(\theta_{43} + \theta_{32} + \theta_{21}) + L_3 \cos(\theta_{32} + \theta_{21}) + L_2 \cos(\theta_{21}))\vec{z}_1\end{aligned}$$

Question 6: Déterminer les conditions permettant de déplacer le point E horizontalement, uniquement suivant $\vec{x}_1$.

$$\begin{cases} \vec{V}(E, 5/0) \cdot \vec{y}_1 = 0 \\ \vec{V}(E, 5/0) \cdot \vec{z}_1 = 0 \end{cases}$$

Condition sur $\vec{z}_1$:

$$\begin{aligned}\vec{V}(E, 5/0) \cdot \vec{z}_1 &= 0 \\ (\dot{\theta}_{43} + \dot{\theta}_{32} + \dot{\theta}_{21})(L_4 + L_5)\vec{y}_4 \cdot \vec{z}_1 + (\dot{\theta}_{32} + \dot{\theta}_{21})L_3\vec{y}_3 \cdot \vec{z}_1 + \dot{\theta}_{21}L_2\vec{y}_2 \cdot \vec{z}_1 \\ &\quad - (\dot{\theta}_{10}(L_4 + L_5) \cos(\theta_{43} + \theta_{32} + \theta_{21}) + \dot{\theta}_{10}L_3 \cos(\theta_{32} + \theta_{21}) \\ &\quad + \dot{\theta}_{10}L_2 \cos(\theta_{21}))\vec{z}_1 \cdot \vec{z}_1 = 0\end{aligned}$$

$$\vec{y}_4 \cdot \vec{z}_1 = \vec{y}_3 \cdot \vec{z}_1 = \vec{y}_2 \cdot \vec{z}_1 = 0$$

$$\begin{aligned}- (\dot{\theta}_{10}(L_4 + L_5) \cos(\theta_{43} + \theta_{32} + \theta_{21}) + \dot{\theta}_{10}L_3 \cos(\theta_{32} + \theta_{21}) + \dot{\theta}_{10}L_2 \cos(\theta_{21})) &= 0 \\ \dot{\theta}_{10}((L_4 + L_5) \cos(\theta_{43} + \theta_{32} + \theta_{21}) + L_3 \cos(\theta_{32} + \theta_{21}) + L_2 \cos(\theta_{21})) &= 0 \\ = \end{aligned}$$

$$f(t)g(t) = 0 \quad \forall t$$

Or, $g(t)$ dépend de la géométrie et n'est pas toujours nul. Donc $f(t) = 0$

$$\dot{\theta}_{10} = 0$$

Condition sur $\vec{y}_1$:

$$\begin{aligned}\vec{V}(E, 5/0) \cdot \vec{y}_1 &= 0 \\ (\dot{\theta}_{43} + \dot{\theta}_{32} + \dot{\theta}_{21})(L_4 + L_5)\vec{y}_4 \cdot \vec{y}_1 + (\dot{\theta}_{32} + \dot{\theta}_{21})L_3\vec{y}_3 \cdot \vec{y}_1 + \dot{\theta}_{21}L_2\vec{y}_2 \cdot \vec{y}_1 \\ &\quad - (\dot{\theta}_{10}(L_4 + L_5) \cos(\theta_{43} + \theta_{32} + \theta_{21}) + \dot{\theta}_{10}L_3 \cos(\theta_{32} + \theta_{21}) \\ &\quad + \dot{\theta}_{10}L_2 \cos(\theta_{21}))\vec{z}_1 \cdot \vec{y}_1 = 0\end{aligned}$$

$$\vec{z}_1 \cdot \vec{y}_1 = 0$$

$$(\dot{\theta}_{43} + \dot{\theta}_{32} + \dot{\theta}_{21})(L_4 + L_5)\vec{y}_4 \cdot \vec{x}_1 + (\dot{\theta}_{32} + \dot{\theta}_{21})L_3\vec{y}_3 \cdot \vec{x}_1 + \dot{\theta}_{21}L_2\vec{y}_2 \cdot \vec{x}_1 = 0$$

$$\begin{aligned}\overrightarrow{y_4} \cdot \overrightarrow{y_1} &= \cos(\theta_{43} + \theta_{32} + \theta_{21}) \\ \overrightarrow{y_3} \cdot \overrightarrow{y_1} &= \cos(\theta_{32} + \theta_{21}) \\ \overrightarrow{y_2} \cdot \overrightarrow{y_1} &= \cos(\theta_{21})\end{aligned}$$

$$\begin{aligned}(\dot{\theta}_{43} + \dot{\theta}_{32} + \dot{\theta}_{21})(L_4 + L_5) \cos(\theta_{43} + \theta_{32} + \theta_{21}) + (\dot{\theta}_{32} + \dot{\theta}_{21})L_3 \cos(\theta_{32} + \theta_{21}) \\ + \dot{\theta}_{21}L_2 \cos(\theta_{21}) = 0\end{aligned}$$

Soit au final :

$$\begin{cases} \dot{\theta}_{10} = 0 \\ (\dot{\theta}_{43} + \dot{\theta}_{32} + \dot{\theta}_{21})(L_4 + L_5) \cos(\theta_{43} + \theta_{32} + \theta_{21}) + (\dot{\theta}_{32} + \dot{\theta}_{21})L_3 \cos(\theta_{32} + \theta_{21}) + \dot{\theta}_{21}L_2 \cos(\theta_{21}) = 0 \end{cases}$$